EAA CHAPTER [___] · [MEETING DATE]

PREHEAT FROM HOME

A cellular Tanis controller you can build yourself — no hangar Wi-Fi, no monthly fee, and it tells you the engine is actually warming.

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THE PROBLEM

The Tanis wants hours. The hangar is 45 minutes away.

6+ hrs

Tanis recommends about six hours or longer of warm-up for full benefit

45 min

Each way, home to the hangar — driving out early just to plug in kills the morning

No Wi-Fi

T-hangar has one duplex outlet and cell signal — nothing else, and I can’t alter the wiring

WHAT'S ON THE SHELF

Plenty of ways to switch an outlet. Few tell you it worked.

Option	Upfront	Ongoing	Confirms it's heating?
HangarBot preheat kit	$729	Monthly data plan	No — switches the outlet
SwitcheOn (sold by Tanis)	$323	$50/yr	No — switches the outlet
Wi-Fi plug + hotspot	~$60–100	Hotspot plan	No — two things to fail
Plug-in timer	~$30	None	No — and Tanis says no timers

Prices: Aircraft Spruce listing (HangarBot), Tanis website Oct 2026 (SwitcheOn); verify before quoting.

WHY I BUILT ONE

I wanted confidence, not just a remote switch

Proof it’s working

Engine and hangar temps on my phone — I see the rise before I leave home

No monthly bill

Particle’s free plan includes the cellular data for a device like this

Built around Tanis rules

On means on — no cycling, no timer, no thermostat on the engine heater

Grows with me

Cabin heat, flood warning and more bolt on later without starting over

THE ONE RULE THAT SHAPES EVERYTHING

The controller switches the outlet. It never thermostats the Tanis.

Tanis systems are self-regulating. Their guide says not to cycle them on and off and not to use timers or thermostats. So “on” stays on until you turn it off — and the system warns you if it’s been on for a day.

THE MINIMUM BUILD

One relay. Two temperature probes. One phone.

iPhone
Shortcuts buttons + push alerts

→

Particle Cloud
Relays commands over cellular

→

Controller
Particle Muon + M404 LTE-M

→

Relay outlet
Switches 120 V to the cord

→

Tanis
Via the oil-door plug

Engine probe
Taped to the Tanis cord, sits inside the cowl

Hangar probe
Ambient reference — the rise is what matters

ARM / SAFE switch
Latching button at the box — kills it locally

WHAT IT DOES

A preheat morning, start to finish

1

Tap “Preheat”

One button toggles it. The controller replies ON or OFF.

2

Watch the rise

“Hangar Status” shows engine, hangar and the difference.

3

Get warned

Push alert if the engine isn’t warming, or Tanis has run 24 h.

4

Unplug and fly

Pull the cord at the oil door — the engine probe comes with it.

If ARM/SAFE is off, the phone says **“relays not active.”** If the controller can’t be reached, it says **“no response”** — never an old reading as current.

THE CLEVER BIT

The engine probe rides on the cord

Nothing is attached to the engine. When you unplug the Tanis cord before flight — which you already have to do — the probe leaves the cowl with it.

- ✓ Can’t be forgotten in the cowling
- ✓ No conflict with CHT or engine-monitor probes
- ✓ Zero changes to the airplane
- ⚠ Reads cowl air, not metal — watch the trend

FAIL-SAFE BY DESIGN

When anything fails, the heat goes off — not on

If this happens…	…the result is
Controller dies or loses power	Relay drops open — outlet off
Cell signal or cloud goes down	Outlet holds its state; phone shows “state unknown”
ARM/SAFE button set to SAFE	Relay can’t energize, in hardware; phone says “relays not active”
Re-armed after SAFE	Stays off until someone sends a new command
Fueling or engine start	Cord is unplugged first — Tanis requires it, and it’s on the checklist

EXPANSION PATH

Start with preheat. Add the rest when you want it.

Add-on	What you add	What you get
Cabin heat	Second relay + cabin probe + ceramic heater	Warm seats, auto shut-off timer
Current sensor	Clamp-on CT on the Tanis circuit	“Confirmed on” in seconds; stuck-relay alarm
Power-loss alert	Small LiPo battery on the controller	Alert when hangar power drops
Flood warning	River-gauge alerts + floor water sensor	Lead time to fly the airplane out
Activity & lights	Motion sensor + light sensor	FBO visit log; lights-left-on alert

The controller has spare inputs for all of these. No rewiring of what you already built.

WHAT IT COSTS

Part	Approx.
Particle Muon kit, North America (M404 LTE-M)	$89
30 A relay module, opto-isolated input	$5
Waterproof DS18B20 probes (2 of 5 pack)	$4
Adafruit DS2484 probe bridge + Qwiic cable	$15
Mean Well 12v DIN rail power supply	$14
8 out DIN rail power distribution	$15
Listed polycarbonate box, receptacle, 15 A cord	$38
Latching lit button, screw-terminal HAT, resistors	$19
12-ga outdoor extension cord (if needed)	$25
2 ft 12-ga outlets	$17
3 conductor 22 awg 50ft	$20
Deutsch Connectors	$10
WAGO DIN rail connector	$15
TOTAL	**$286**

Kit and bridge prices from vendor listings, Oct 2026.

Estimated total

~$286

Monthly

$0

Particle Free Plan: up to 100 devices, 100K data operations/month, cellular data included

WHAT IT TAKES TO BUILD ONE

If you can wire a panel, you can build this

Skills

One relay module, one receptacle in a listed box, screw terminals and a multimeter. Simpler than any panel you've wired.

Software

No coding. Set up the device in a browser, flash my shared firmware, install iPhone Shortcuts.

Time

An evening on the bench, an hour at the hangar, then a 24–48 hour soak test before you trust it.

You'll need: an iPhone, cell signal inside your hangar (we test that first), and a Tanis or similar plug-in preheater.

INSIDE THE BOX

Four details that make it fail safe

Load on COM → NO, never NC

A dead relay means a dead outlet. Wired through NC, it would heat whenever the controller is off.

Active-high + 15 kΩ pull-down

In active-low mode the outlet came on at power-up. Active-high keeps it off before firmware runs.

ARM/SAFE cuts relay power

The latching button is in the relay’s 5 V feed. SAFE means no coil power, whatever the code does.

Listed plastic enclosure

Contains a 120 V fault near fuel, and doesn’t block the cellular antenna. Magnets, no drilling.

WHERE IT STANDS TODAY

Honest status: working on the bench, not yet in service

Done and tested

- Cellular link inside the hangar
- Both relays: on/off, power-up, reset, OTA update
- ARM/SAFE cuts relays in hardware
- One-tap iPhone toggle, confirmed by the controller

Still to do

- Temperature probes — bridge part on order
- Engine probe at full cable length
- Status + alert firmware, shared Shortcuts
- 24–48 hour soak test at the hangar

Builder’s guide goes out once these are closed — target: **[date]**.

WANT TO BUILD ONE?

Here’s how we get there together

1

Sign up tonight

Name, hangar location, preheater model

2

Signal check

I bring my controller to your hangar

3

Group order

Once the guide is released

4

Build night

Bench builds together at [location]

Questions?